SipChill: System Engineering Architecture & Technical Feasibility Report

Design Trade-offs, Thermal Physics, Embedded Stack, and Deprecated Approaches

Abstract

This technical report describes the engineering architecture, mathematical modeling, and design iterations of **SipChill**, an active thermal regulation cup for beverages designed using Internet-of-Things (IoT) technologies. This report covers system level requirements, thermal dissipation models, integration of thermoelectric cooling (TEC), fluid boundary layer reduction, power stage controllers, and firmware architecture. In addition to that, this report lists out all the eliminated architectures (such as microcontrollers, mechanical stirring mechanisms, raw air cooling, and bio-sensing telemetry systems) with reasoning behind eliminating them.

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1 System Overview & Commercial Objectives

SipChill was made in such a way that it can be used as an active solid-state consumer cooling solution that would help keep the products cool and precisely at set points. While passive solutions rely on vacuum-insulated flasks, SipChill uses an active thermoelectric heat-pumping engine.

1.1 Performance Targets

- **Beverage Volume:** 350–500 mL standard volume capacity.
- **Thermal Pull-down:** Active temperature reduction of liquid by ≥ 15°C within 12–18 minutes.
- **Steady-State Maintenance:** Sustained liquid holding temperature (4°C to 10°C) under ambient loads (25°C to 35°C).
- **Form Factor:** Handheld ergonomics, decoupled washability, zero food-zone contamination.

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2 Thermal Dynamics & Physics Principles

2.1 Thermoelectric Cooler (TEC) Governing Equations

The thermal engine relies on the Peltier effect across bismuth telluride (Bi_2Te_3) semiconductor junctions. The net rate of heat absorbed at the cold junction (Q_c) is governed by:

$$Q_c = S \cdot I \cdot T_c - \frac{1}{2} I^2 R - K \cdot (T_h - T_c) \tag{1}$$

Where:

- S = Seebeck coefficient (V/K)
- I = Operating DC current (A)
- T_c, T_h = Absolute temperatures of the cold and hot junctions (K)
- R = Internal electrical resistance of the module (Ω)
- K = Thermal conductance across module pellets (W/K)

The coefficient of performance (COP) for the cooling cycle is defined as:

$$\text{COP} = \frac{Q_c}{P_{\text{electrical}}} = \frac{S \cdot I \cdot T_c - \frac{1}{2} I^2 R - K \Delta T}{I^2 R + S \cdot I \cdot \Delta T} \tag{2}$$

Given that the COP is typically low ($0.3 \leq \text{COP} \leq 0.6$), the hot junction heat rejection rate (Q_h) significantly exceeds the cold side intake:

$$Q_h = Q_c + P_{\text{electrical}} \tag{3}$$

For a 30 W cooling rate (Q_c) operating at an electrical input of 50 W, the dissipation system must evacuate upwards of 80 W of continuous thermal flux without elevating T_h.

2.2 Phase Change Material (PCM) Thermal Buffering

To eliminate the requirement for massive, high-decibel forced-air heatsinks during the transient pull-down phase, a latent heat thermal energy storage (LHTES) buffer wraps the hot-side dissipation block.

The heat absorbed during the phase transition of the PCM is calculated via:

$$Q_{\text{latent}} = m \cdot \Delta H_f \tag{4}$$

Where m is the mass of the PCM and ΔH_f is the specific latent heat of fusion (J/kg).

- **Material Selection:** Paraffin waxes or salt-hydrate compounds with a phase transition point between $40°\text{C} \leq T_{\text{phase}} \leq 45°\text{C}$.
- **Functional Mechanism:** During peak flash cooling, high heat flux from Q_h is absorbed isothermally by the PCM melting cycle. This prevents T_h from runaway saturation, keeping $\Delta T = (T_h - T_c)$ low and maximizing TEC electrical efficiency.
- **Regeneration:** Once the liquid reaches setpoint and TEC power throttles down, the heat stored in the PCM is slowly evacuated to ambient via low-RPM boundary conduction.

2.3 Fluid Stratification and Boundary Layer Transport

In an unagitated fluid vessel, dense cold liquid settles at the conductive interface at the bottom ($T \approx 4°\text{C}$), while buoyant warm liquid remains elevated. This creates an insulating thermal boundary layer:

$$q'' = -k_{\text{fluid}} \left. \frac{\partial T}{\partial z} \right|_{z=0} \tag{5}$$

Because stationary water has low thermal conductivity ($k \approx 0.6\,\text{W}/(\text{m} \cdot \text{K})$), the conductive cold base rapidly approaches freezing, triggering condensation/frost and stalling convective heat transport from the core liquid volume.

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3 System Hardware Architecture

3.1 Physical Subsystem Partitioning

To ensure user safety, regulatory ingress protection (IPX7), and dishwasher compatibility, the architecture is split into decoupled sub-assemblies:

1. **Base Power & Thermal Engine:** Contains the TEC, PCM chamber, finned heat sinks, low-noise micro-blower, multi-cell battery system, and the primary high-power PCB.

2. **Vacuum Vessel & Interface:** Double-wall stainless steel body with a vacuum boundary surrounding the lateral walls ($\leq 10^{-3}\,\text{Torr}$) to nullify ambient convective and conductive parasitic load. The bottom comprises a clad copper-aluminum conduction plate that interfaces directly with the cold plate of the base module.

3. **Smart Cap Module:** Houses a low-power microcontroller/BLE peripheral, capacitive top-ring interface, miniature OLED/LED array for temperature display, and liquid consumption tracking.

3.2 Fluid Agitation: Ultrasonic Acoustic Streaming

To disrupt thermal stratification without mechanical rotating shafts, a high-frequency piezoelectric transducer is coupled to the conductive bottom base.

- **Operating Principle:** Ultrasonic excitation (28 kHz to 40 kHz) induces non-linear Reynolds stress in the acoustic boundary layer, producing steady bulk convective flow (acoustic streaming).
- **Engineering Advantages:** Completely solid-state, zero food-contact contamination, no micro-fissures from rotating seals, silent operation, and immediate destruction of the conductive base thermal boundary layer.

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4 Electrical & Firmware Design

4.1 Compute Framework

The core logic is executed on an **ESP32-S3** system-on-chip, leveraging:

- Dual-core Xtensa 32-bit LX7 architecture running FreeRTOS. Core 0 manages the sensor capture and high-frequency closed-loop control tasks; Core 1 handles BLE communication and display peripherals.
- Integrated hardware PWM generators for precise power conversion.
- Native low-power modes during quiescent holding periods.

4.2 Power Electronics & Battery Management

- **Energy Storage:** (11.1 V nominal, 12.6 V peak) high-discharge 18650 or 21700 Lithium-Ion cells (rated ≥ 15 A continuous discharge).
- **Power Switching:** Dedicated low-side gate driver driving logic-level N-channel MOSFETs with ultra-low on-resistance ($R_{DS(\text{on})} \leq 3.5\,\text{m}\Omega$).
- **Inductive LC Filtering:** Pure PWM switching of Peltier modules causes internal temperature swings that degrade efficiency. A low-pass LC smoothing filter provides continuous DC output across the TEC terminals.
- **Charging/Sinking:** USB-C Power Delivery (PD 3.0 / EPR) sink controller (e.g., CH224K) to negotiate 12 V or 20 V profiles up to 65 W directly from off-the-shelf power bricks.

4.3 Control Theory: Anti-Windup PID Implementation

Liquid temperature feedback is continuously computed via an internal NTC thermistor array. The output drives a discrete PID loop:

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau)d\tau + K_d \frac{de(t)}{dt} \tag{6}$$

To prevent integrator saturation during the high-power pull-down phase, back-calculation anti-windup clamping is implemented:

$$e(t) = T_{\text{set}} - T_{\text{actual}} \tag{7}$$

When the actuator output hits the upper duty cycle bound ($D_{\max} = 100\%$), integration is conditionally halted until the liquid enters the proportional band ($\pm 1.5^\circ$C).

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5 Comparative Architecture & Deprecated Paths

Table 1 outlines the architectural concepts explored during the research and development phase, detailing the design directions selected versus those permanently discarded.

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6 Summary & Immediate Next Steps

The engineering path for SipChill prioritizes solid-state reliability, thermal efficiency, and commercial safety over hobbyist shortcuts:

1. Finalize CAD integration for the thermal stack: cold-plate interface, TEC module, PCM encapsulation chamber, and heat pipe routing.

Subsystem	Evaluated Option	Decision	Engineering Justification
Processing			
	ESP32 Platform	**Adopted**	Dual-core processing, hardware PWM, integrated BLE/Wi-Fi stack, robust FreeRTOS support.
Agitation	Magnetic Stirrer	**Discarded**	Dynamic friction seals introduce leak vectors; motor bearings cause acoustic whine; hygiene and cleaning hazards in beverage vessels.
	Ultrasonic Transducer	**Adopted**	Solid-state acoustic streaming, no moving components, reliable fluid boundary layer disruption.
Thermal	Pure Air-Cooling	**Discarded**	High steady-state thermal resistance causes immediate thermal runaway on high-current TEC modules in compact volumes.
	Hybrid PCM + Air	**Adopted**	Acts as a thermal capacitance stage; absorbs transient heat spikes during maximum pull-down phases.
Feature Set			
	Precision Thermal Control	**Adopted**	Focused strictly on rapid cooling, steady-state thermal maintenance, and battery runtime.
Structure	Monolithic Body	**Discarded**	Heavy, unergonomic, impossible to submerge/wash the fluid container safely.
	Decoupled Base/Vessel	**Adopted**	Allows dishwasher-safe container while isolating electronics and power stages in a dry base unit.

Table 1: System Design Trade-offs and Architectural Decisions

2. Fabricate the unified custom PCB: ESP32-S3 microcontroller, USB-C PD sink stage, discrete LC filter, and low-$R_{DS(\text{on})}$ MOSFET driver.
3. Implement and tune the anti-windup PID controller with secondary thermal runaway cutoff triggers for hardware protection.